

Part 2: Build an LLM Agent with Gentopia [50 pts + 20 bonus pts]

In this part, we will switch to an extended topic called “LLM agents”. In Part 1, we have mainly used gpt-4o-mini as a question-answer system, but an LLM can be formulated to be a “vivid” agent who learns to use tools and helps us in broader tasks, just like a virtual assistant!

To this end, we will be using an open-source LLM agent implementation framework, called “Gentopia” [2]. The original code repository can be found at <https://github.com/Gentopia-AI>, but note that for this assignment, we will use this adapted “Gentopia-Mason” version at <https://github.com/LittleYUYU/Gentopia-Mason/tree/main>.

Prerequisite: Library Installation

As the first step, please make sure to install all required Python packages following the instructions on README. More specifically, it means to run the following lines of code:

```

1 # Clone the repository
2 git clone git@github.com:LittleYUYU/Gentopia-Mason.git
3 cd Gentopia-Mason
4
5 # Create a conda virtual environment
6 conda create --name gentenv python=3.10
7 conda activate gentenv
8 pip install -r requirements.txt
9
10 # Set up global environment
11 export PYTHONPATH="$PWD/Gentopia:$PYTHONPATH"
12
13 # Set up OpenAI API key
14 cd GentPool
15 touch .env
16 echo "OPENAI_API_KEY=<your_openai_api_key>" >> .env

```

Task 3: Build a Vanilla LLM Agent

In this task, you will follow the instructions at <https://github.com/LittleYUYU/Gentopia-Mason/tree/main?tab=readme-ov-file#quick-start-clone-a-vanilla-llm-agent> to create a vanilla LLM agent. This agent is essentially the same question-answer model you have tried in Part 1.

The agent, under the framework of Gentopia, is configured as follows (with annotations):

```

1 # ./GentPool/gentpool/pool/<your_agent_name>/agent.yaml
2 # Vanilla agent template
3
4 name: <your_agent_name>
5 version: 0.0.1
6 type: vanilla
7 description: A plain gpt-4 LLM. # this is a brief description of the agent
8 target_tasks:
9   - anything to do with an LLM
10 llm:
11   model_name: gpt-4 # indicating the API version
12   params:
13     temperature: 0.0 # both temperature and top_p are used to control the
14       sampling diversity

```

Task 4: Build a Tool-Augmented LLM Agent

Now, we will follow the instructions at <https://github.com/LittleYUYU/Gentopia-Mason/tree/main?tab=readme-ov-file#implement-a-scholar-llm-agent-with-tool-augmentation> and create a tool-augmented LLM agent. In particular, this agent, called a “scholar agent”, has been equipped with tools to access Google Scholar.²

This tool use does not come for free; it relies on Python implementation using the “scholarly” library,³ which can be found within the gentopia source code at https://github.com/LittleYUYU/Gentopia-Mason/blob/main/Gentopia/gentopia/tools/google_scholar.py.

Again, **change the “model_name” field to “gpt-4o-mini”**.

10. (10 points) Now, let’s the same query for Prof. Ziyu Yao’s papers again:

User: Find papers written by Ziyu Yao at George Mason University
agentrinkischolar:

Here are some papers written by Ziyu Yao at George Mason University: 1 Unifiedskg: Unifying and multi-tasking structured knowledge grounding with text-to-text language models • Publication Year: 2022 • Venue: arXiv preprint arXiv:2201.05966 • Citations: 278 2 CoaCor: Code Annotation for Code Retrieval with Reinforcement Learning • Publication Year: 2019 • Venue: The Web Conference 2019 • Citations: 117 3 Staqc: A systematically mined question-code dataset from stack overflow • Publication Year: 2018 • Venue: Proceedings of the 2018 World Wide Web Conference • Citations: 102 4 Model-based interactive semantic parsing: A unified formulation and a text-to-SQL case study • Publication Year: 2019 • Venue: 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP’19) • Citations: 75 5 Reinforced dynamic reasoning for conversational question generation • Publication Year: 2019 • Venue: arXiv preprint arXiv:1907.12667 • Citations: 55

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Does the response look reasonable now? Based on the meta information that popped out through the agent’s responding process, can you describe how the agent made it right this time?

It looks like the agent fetched information on Ziyu Yao first. Then queried to find the top 5 papers for Ziyu Yao using an id for this professor it fetched. Then the agent returned title, publication year, citations, etc. It was now able to query a database or page with the information pertaining to Ziyu Yao and their publications.

11. (5 points) Look at the configuration file of your scholar agent at `./GentPool/gentpool/pool/<your_agent_name>/agent.yaml` and see what other functions are supported. Try a few other

²<https://scholar.google.com/>

³<https://pypi.org/project/scholarly/>

things you found interesting! Some examples to consider: Can you summarize the paper titled "Gentopia: A collaborative platform for tool-augmented llms"?, What papers have cited "Gentopia: A collaborative platform for tool-augmented llms", etc.

User: How many people have cited Ziyu Yao?

your_agent_name:

'name': 'Ziyu Yao', 'uid': '4IYrMNUAAAAJ', 'affiliation': 'Assistant Professor, George Mason University', 'interests': 'Natural Language Processing', 'LLMs', 'Human-AI Interaction', 'Question Answering', 'Code Generation', 'citation': 973

'name': 'Ziyu Yao', 'uid': '3eePDdIAAAAJ', 'affiliation': 'Peking University', 'interests': 'Generative AI', 'Multimodal Large Language Model', 'citation': 57
There are two authors named Ziyu Yao: 1 Ziyu Yao (Assistant Professor, George Mason University) has been cited 973 times. 2 Ziyu Yao (Peking University) has been cited 57 times. If you need information about a specific one, please let me know!

"name": "search_author_by_name", "arguments": "author": "Ziyu Yao", "top_k": 5

'name': 'Ziyu Yao', 'uid': '4IYrMNUAAAAJ', 'affiliation': 'Assistant Professor, George Mason University', 'interests': 'Natural Language Processing', 'LLMs', 'Human-AI Interaction', 'Question Answering', 'Code Generation', 'citation': 973

'name': 'Ziyu Yao', 'uid': '3eePDdIAAAAJ', 'affiliation': 'Peking University', 'interests': 'Generative AI', 'Multimodal Large Language Model', 'citation': 57
There are two authors named Ziyu Yao: 1 Ziyu Yao (Assistant Professor, George Mason University) has been cited 973 times. 2 Ziyu Yao (Peking University) has been cited 57 times. If you need information about a specific one, please let me know!

12. (15 points) *You may have realized that the current scholar agent still fails to answer general questions such as “Tell me something about George Mason NLP group”, though they are also relevant to “scholar”. In addition, it cannot read PDFs for summarizing papers either, so it cannot provide answers to specific questions about the paper details. Can you figure out a way for the agent to do Google Search (5 pts) and PDF reading (10 pts), and present a few examples below? (Hint: Google Search has been provided in Gentopia’s source code, whereas you could opt for any Python tools you like, e.g., PyPDF2,⁴ for PDF reading.)*

User: Can you summarize the pdf <https://arxiv.org/pdf/2407.02067>

agentrinkischolar: The paper titled “Crossroads of Continents: Automated Artifact Extraction for Cultural Adaptation with Large Multimodal Models” by Anjishnu Mukherjee, Ziwei Zhu, and Antonios Anastasopoulos presents a comprehensive study on the cultural awareness of large multimodal models (LMMs). The study is structured into three main phases: 1 Cultural Context Recognition: The authors introduce a dataset called DALLE STREET, which consists of 9,935 images representing 67 countries and 10 cultural concept classes. This dataset is generated using DALL-E 3 and validated by human annotators. The study reveals disparities in cultural understanding at the sub-region level when tested with both open-weight (LLaVA) and closed-source (GPT-4V) models. 2 Artifact Extraction: The second phase involves an artifact extraction task where over 18,000 cultural artifacts associated with different countries are identified. This task aims to assess the models’ deeper understanding of cultural contexts by extracting implicit cultural artifacts from images. 3 Cultural Adaptation Pipeline: The authors propose a pipeline called CULTURE ADAPT, which allows for the adaptation of images from one cultural context to another. This pipeline utilizes the insights gained from the artifact extraction task to modify images while maintaining their original structure. The findings highlight the nuanced cultural competence of LMMs and emphasize the need for developing culture-aware systems. The study also discusses the implications of cultural biases in LMMs and suggests that future research should focus on improving cultural representation in AI systems. Overall, the paper contributes to the understanding of how LMMs can recognize, represent, and adapt cultural content, providing a foundational benchmark for future research in this area..

User: Can you tell me about the University of Richmond?

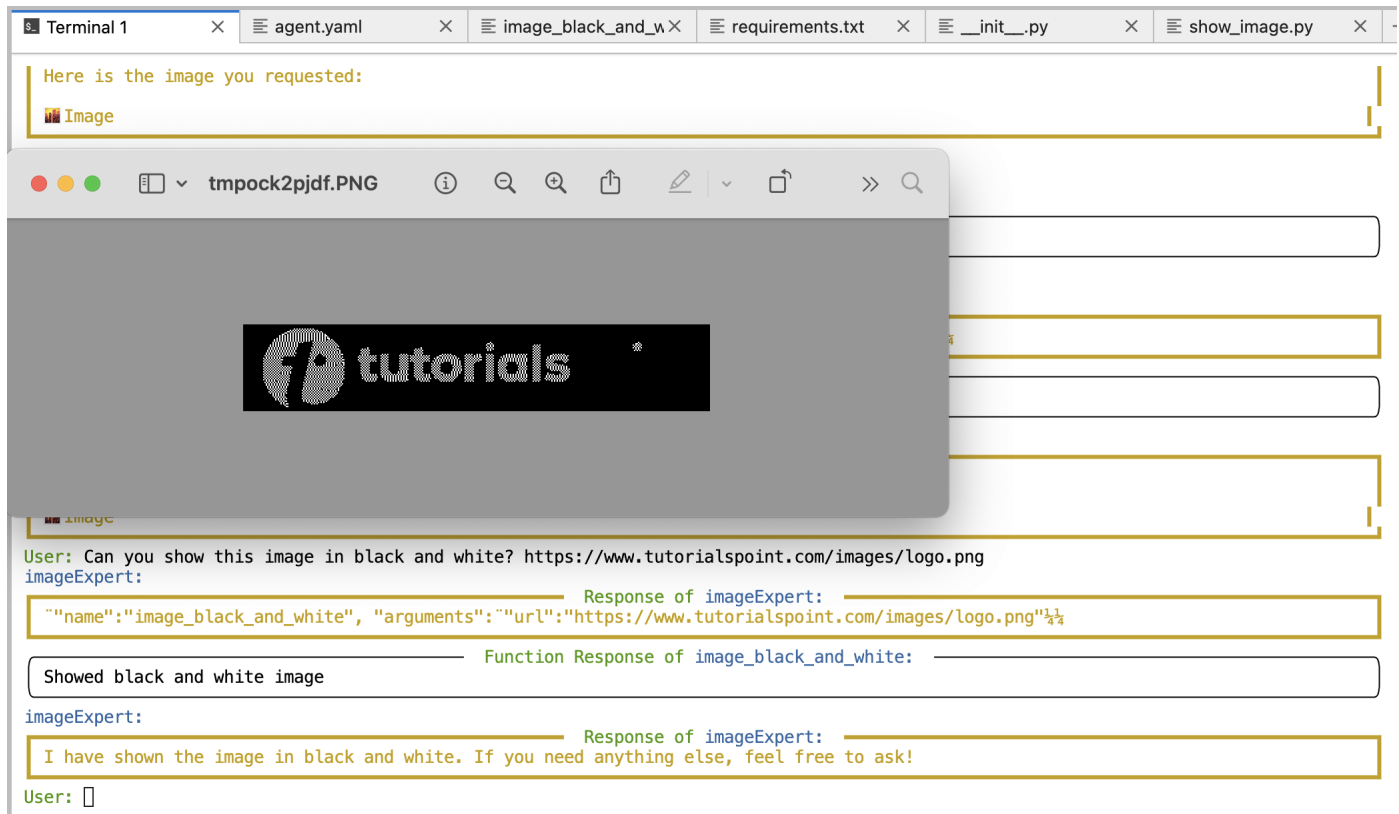
agentrinkischolar: The University of Richmond is a highly regarded liberal arts university located in Richmond, Virginia. It offers a unique combination of liberal arts education alongside programs in law, business, and leadership. The university is known for its interdisciplinary approach, integrating various fields of study to provide a comprehensive educational experience. Key features of the University of Richmond include: • Academic Excellence: The university is recognized for its rigorous academic programs and exceptional student services. • Campus: It boasts a beautiful campus that enhances the student experience. • Athletics: The university’s athletic teams are known as the Spiders, and they compete in various sports at the collegiate level. For more detailed information, you can visit their official website: University of Richmond.

Bonus: Build Your Own Agent! (20 pts)

We will provide up to 20 points to students who implement a different tool-augmented LLM agent for positive use (i.e., the agent cannot be designed for unethical purposes). To claim bonus points, the agent needs to be substantially different from the scholar agent and others included in Gentopia’s agent pool. Be creative! Partial credits may be provided based on student effort.

I implemented an agent called imageExpert to show images from urls. Then, you can also give a url and

⁴<https://www.geeksforgeeks.org/working-with-pdf-files-in-python/>



Acknowledgment

The mathematical problems used in this assignment come from the GSM8k dataset [1], and the source code of Gentopia comes from the referred EMNLP'23 paper [2].

References

- [1] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [2] Binfeng Xu, Xukun Liu, Hua Shen, Zeyu Han, Yuhao Li, Murong Yue, Zhiyuan Peng, Yuchen Liu, Ziyu Yao, and Dongkuan Xu. Gentopia.AI: A collaborative platform for tool-augmented llms. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 237–245, 2023.